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QUALITY PAPER

Smart technologies for collection and classification of electronic waste

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Abstract

Purpose – The main aim of this study is to understand the role of smart technologies and show the rankings of various smart technologies in collection and classification of electronic waste (e-waste).

Design/methodology/approach – This study presents a framework integrating the concepts of collection and classification mechanisms and smart technologies. The criteria set includes three main, which are economic, social and environmental criteria, including a total of 15 subcriteria. Smart technologies identified in this study were robotics, multiagent systems, autonomous tools, smart vehicles, data-driven technologies, Internet of things (IOT), cloud computing and big data analytics. The weights of all criteria were found using fuzzy analytic network process (ANP), and the scores of smart technologies which were useful for collection and classification of e-waste were calculated using fuzzy VlseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR).

Findings – The most important criterion was found as collection cost, followed by pollution prevention and control, storage/holding cost and greenhouse gas emissions in collection and classification of e-waste. Autonomous tools were found as the best smart technology for collection and classification of e-waste, followed by robotics and smart vehicles.

Originality/value – The originality of the study is to propose a framework, which integrates the collection and classification of e-waste and smart technologies.

Keywords Electronic waste, Collection and classification, Smart technologies, Fuzzy ANP, Fuzzy VIKOR

Paper type Research paper

1. Introduction

Since the consumption of electrical and electronic equipment increases tremendously for various reasons (Singh *et al.*, 2022), electronic waste (e-waste) became as one of the most critical problems (Shittu *et al.*, 2021). For instance, only 20% of 45 million tons of e-waste was collected in 2016 (Baldé *et al.*, 2017). The amount of e-waste is also growing very fast (Wang *et al.*, 2021; Shahabuddin *et al.*, 2022). The total amount of e-waste increases at the rate of 2.6 million annually (Shreyas Madhav *et al.*, 2022), and it is projected to reach 75 million tons in 2030 (Mohammadi *et al.*, 2021).

In emerging economies, e-waste is a more critical issue compared to developed ones (Ikhlayel, 2018) in consequence of lack of awareness to waste management practices (Kumar *et al.*, 2017). Emerging economies just have policies for collection and classification of e-waste,

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and therefore, there is a need for a framework to standardize the routine regulations to improve the effectiveness of e-waste management mechanisms (Sagnak *et al.*, 2021).

In particular, e-waste increases the greenhouse gases; therefore, a need arises for new sustainable practices in e-waste management. However, achieving sustainable practices for the e-waste management is complicated because of industry-specific incidents such as concentrated nonhealthy metal usage, heavy energy usage, intensive chemical process and slow-building stock turnover (Joensuu *et al.*, 2020). Besides, e-waste recycling became more difficult because of significant differences in the price of recovered materials and technological barriers (Copani *et al.*, 2019). Unlike other waste, a significant effort is required to collect and sort out e-waste due to the variety of materials being used in compact form of electronic products, which is one of the significant challenges of e-waste recycling (Sahajwalla and Hossain, 2020).

Various researchers (Dave *et al.*, 2016; Garlapati, 2016; Widmer *et al.*, 2005; Robinson, 2009) stated that e-waste comprises several items and is a piece of equipment operating with a cord or battery (e.g. household electronics, information technology and telecommunication devices, consumer devices, healthcare devices, lightning and illumination equipment items, tools and machinery, monitoring and control equipment items, automatic dispenser machines, entertainment and athletics).

All electronic products, which have expired, and cannot be restructured, are stated as e-waste (Widmer *et al.*, 2005). E-waste differs from other forms of wastes due to costly and dramatic environmental impacts. Numerous emerging economies are grappling with a large amount of locally generated or imported e-waste (Ikhlayel, 2018). Turkey, which can be claimed as an emerging economy, has been facing with many e-waste controlling problems (Baldé *et al.*, 2017).

In this respect, it is necessary to propose a framework and guideline, and develop a roadmap to handle e-waste problems is emerging economies. The first research question is therefore established as:

RQ1. How can a framework be proposed to develop a roadmap for e-waste collection and classification?

Also, the spread of information technologies introduced in the past couple of years, and the digitalization of world society had a significant effect on satisfying people's needs at the cost of increasing pollution and degradation (Archibong Bassey and Roseline William, 2020). As Bjorn *et al.* (2020) stated, the detractive effect of new technologies on the environment has come researchers' attention. Green information technology (IT) indicates the relationship between information technologies and environmental studies. According to Feenberg (2019), its main objective has been recognized that digital technologies support industrial improvements and provide a sustainable environment for people (Broo *et al.*, 2021).

Within this perspective, there is a significant gap in the past studies with respect to the integration of smart technologies with e-waste collection and classification mechanism (Rada *et al.*, 2013). Adaptation of smart technologies has become important in the e-waste operations (Kazancoglu *et al.*, 2021). Therefore, this study presents a framework integrating the concepts of collection and classification mechanisms and smart technologies. The motivation of the study is to integrate the benefits of smart technologies to the collection and classification of e-waste. The uniqueness of the study is to propose a framework, which integrates the collection and classification of e-waste and smart technologies. The criteria set was developed for the different characteristics of collection and classification centers, and the smart technologies were identified as the solutions for collection and classification mechanism. The criteria set include three main, which are economic, social and environmental criteria, including a total of 15 subcriteria. Smart technologies identified in this study were robotics, multiagent systems, autonomous tools, smart vehicles, data-driven technologies, Internet of

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things (IOT), cloud computing and big data analytics. The weights of all criteria were found using fuzzy ANP, and the scores of smart technologies which were useful for collection and classification of e-waste were calculated using fuzzy VlseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR). The implementation was conducted in ten electronics companies, which create lots of e-waste, in Izmir, Turkey.

In this respect, the other research questions are established as:

RQ2. Which criteria list is critical for collection and classification mechanism for e-waste?

RQ3. Which smart technologies can be useful for collection and classification of e-waste?

RQ4. How can the criteria weighting and smart technologies ranking be calculated?

The structure of study is as follows: Section 2 highlights the theoretical background for collection and classification of e-waste and the use of smart technologies. Sections 3 and 4 present the proposed framework, and methodology, respectively. Section 5 summarizes the case study, and shows the findings. Section 6 demonstrates the implications and finally, Section 7 shows the concluding remarks, limitations,and possible future directions.

2. Theoretical background

It is understood from past studies that there is a gap in the e-waste collection and classification mechanism, particularly in the track-and-trace technology and smart collection systems-based collecting process (Lopez Alvarez *et al.*, 2008; Rada *et al.*, 2013; Keim, 2017; Martin and Leurent, 2017). Additionally, digitization in the collection process should be planned to use support and synchronize shifts toward sustainability (Keim, 2017). Hence, embracing innovations and using appropriate technology have become crucial under the triple bottom line (TBL) concept in order to generate innovations in the area of waste management and to embrace the digital age (Osburg and Lohrmann, 2017).

This section shows the research areas of collection and classification centers for e-waste and smart technologies.

2.1 E-waste collection and classification problem

E-waste is most critical form of waste in emerging economies (Ikhlayel, 2018). This has two reasons. First, e-waste is locally-created and cannot be properly destroyed (Park *et al.*, 2017), and second, it was imported from developed countries for disposal (Garlapati, 2016; Chandra *et al.*, 2022). In fact, for emerging economies, this creates an opportunity to get usable materials by reprocessing the e-waste; but, e-waste also cannot be properly reprocessed (Heeks *et al.*, 2015). Inappropriate reprocessing, recovery, recycling, reusing or remanufacturing of the materials have unfavorable effects on human health, air, water and soil, which means that inappropriate disposal and nonclassification of e-waste results in lots of hazardous consequences (Park *et al.*, 2017).

The new technologies are sometimes used excessively in the electronic goods manufacturing, and creates huge amount of e-waste. Therefore, the most critical activity is to develop efficient collection and classification mechanism (Garlapati, 2016). There is a lack of waste management system in emerging economies, in which there is also a lack of regulations for collection, classification, disposal of e-waste (Mundada *et al.*, 2004).

E-waste is stored because of the lack of proper collection, classification and disposal systems (Ikhlayel, 2018). There are concerns about the separation, disposal and recycling of e-wastes. In such economies, concerns are based on the huge imported volumes of e-waste, lack of regulations and the informal e-waste reprocessing activities, which was not promoted by the local governments (Gu *et al.*, 2016). Moreover, e-waste is collected, classified and disposed

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illegally (Ikhlayel, 2018). In addition, the collection process is the responsibility of unskilled and uneducated workforce in such economies (Kumar *et al.*, 2017). If total e-waste both produced and imported was not appropriate for reprocessing activities, then these were burned, which results in environmental damage and negatively affects human health (Kumar *et al.*, 2017).

Nowadays, the life cycle of electronic products was shortened in consequence of the rapid developments in industry. This increases the e-waste amount. In Turkey, in 2012, a regulation was coded to determine the technical and legal guidelines for reducing the e-waste through recycling, remanufacturing and reusing activities (Ozturk, 2015). The main aim of this regulation was to decrease and control the e-waste. With this regulation, the municipalities and enterprises were obligated to collection, processing and disposal of e-waste. However, although the collection and classification mechanism was determined through regulation, Turkey still have huge e-waste problem. The collection and classification system cannot be implemented effectively by the enterprises. E-waste is collected; however, they are not classified (Park *et al.*, 2017).

2.2 Smart technologies

Mechanical systems incorporating actuators, sensors and preprogrammed controllers that enable a machine to adjust to erratic environmental circumstances are included in the category of "smart technologies." It necessitates understanding of the mechanical system, embedded sensors, controlled devices, driving electronics and integrated software that gives the system intelligence. The application of the multidisciplinary principles of smart technologies has been significantly impacted by the convergence of machine and computational technology advancements. The main concept is to create an integrated technology made up of the components, which are: (1) a distributed sensing system capable of tracking the machine's response; (2) actuators capable of altering the machine's behavioral characteristics and (3) control units capable of executing a preplanned strategy.

According to Jahangir *et al.* (2021), a variety of intricate factors, including advancements in digitization, the condition of electrical components, network infrastructure, the development of sensors and software solutions affected the development of smart technologies. Safe energy is another crucial component of smart technologies, which call for robotic instruments, a huge number of sensing tools, data centers, server stations, etc. This implies that in the future, renewable energy technologies will be able to meet energy demands for the manufacture of electronics.

Implementing smart technologies in the circular activities is reportedly promising to invest in and introduce fresh ideas to effective e-waste management (Zhou *et al.*, 2021). According to Ashwin *et al.* (2021), there are a number of IoT-enabled solutions for smart bin creation, smart e-waste segregation and smart trash truck management. According to Murugesan *et al.* (2021), there will soon be a variety of ways for managing e-waste.

In order to decrease transportation and material sorting costs, Alcayaga *et al.* (2019) studied smart e-waste collecting systems based on IoT. The goal of the Fatimah *et al.* (2020) study is to create an industry 4.0-based national e-waste management system.

For a more circular economy, smart technologies can motivate connected industries to enhance e-waste management procedures. However, there are still many restrictions and difficulties (Anagnostopoulos *et al.*, 2017; Zhang *et al.*, 2019a; Sharma *et al.*, 2020). According to Fuss *et al.* (2018), e-waste management system implementation is still in its early stages. Twelve major barriers to smart waste management are presented by Zhang *et al.* (2019a), including a lack of knowledge about smart waste management, innovation capacity, challenges with technologies and their applications, regulatory pressures, market pressures and demands, a lack of stakeholder cooperation, cost and financial challenges, a lack of environmental protection and education, and the pursuit of profits over environmental

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protection, lack of cluster effect, lack of leadership commitment, and lack of proper standards of waste management.

Sharma *et al.* (2020) created a framework for IoT adoption in the e-waste management systems by identifying the most significant IoT barriers as a lack of standards, legislation and Internet access. All smart e-waste management systems, according to Borghi *et al.* (2014), should have a planning strategy for efficiency within a circular economy framework.

Lu *et al.* (2017) proposed radio-frequency identification (RFID)-based multistage e-waste collection and classification systems to increase the effectiveness of e-waste collection. Rutqvist *et al.* (2019) used automated machine learning techniques to overcome the e-waste classification problem using specific sensors. Imran *et al.* (2017) presented a waste management system to measure and clear the containers. Predictive and descriptive data analysis approaches are used for effective e-waste management planning. Except for data visualization in geographic information system (GIS), visual analysis of e-waste can be used for e-waste collection and classification.

From algorithmic perspective, various e-waste classification models based on some powerful algorithms like artificial neural network (ANN) (Hoque and Rahman, 2020) or convolutional neural network (CNN) (Ahmad *et al.*, 2021) were used as well as a multilayer perceptron, *k*-nearest neighbors (KNN) algorithms, gradient boosting tree and simulation approaches.

Cong *et al.* (2021) presented an IoT and deep learning-based smart waste management system to classify, monitor and collect solid waste and other recyclable waste. Xia *et al.* (2021) employed linear models to predict input picture components.

The key issue is how to control e-waste using such information technologies or methods. An answer to the question can be in fields such as multiagent systems, robotics, autonomous tools, digital twins, image processing, smart vehicles, pattern recognition, wireless communication and data-driven technologies. The process of digitization will also redefine what "waste" is, as well as offer driverless collecting methods and robotic solutions (Martin and Leurent, 2017).

Problems with cost and time optimization, in particular, are brought on by the lack of use of digital technologies in the waste collecting process (Yukalang *et al.*, 2017). Manufacturing, inventory, collection, classification, processing, recovery and burning are all aspects of managing e-waste. By integrating digital technology into the e-waste collecting process, money and time spent each collation station may be recovered and the process is traceable (Rada *et al.*, 2013).

Industries are becoming more and more interested in smart technologies used for designing and creating solutions to problems with waste management, such as e-waste collection, transportation, classification and sorting (Shevchenko *et al.*, 2021). The development of communication and data transmission technologies, field experiment technologies and truck route planning and scheduling technologies are among the categories of smart technologies that the waste management systems are divided into (Esmaeilian *et al.*, 2018).

3. Proposed framework

This section presents the novel framework to demonstrate the flow of the study for smart technologies using for collection and classification of e-waste. The criteria list consists of three main criteria; economic, social and environmental criteria, including a total of 15 subcriteria. Smart technologies include eight different alternatives including robotics, multiagent systems, autonomous tools, smart vehicles, data-driven technologies, cloud computing, IOT and big data analytics.

After an intensive literature review, the proposed 15 criteria were validated through four academics and three industrial experts. The academic experts include professors from business administration, logistics, information management and management information system departments from various universities. The industrial experts are supply chain managers,

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supply chain vice managers and sustainability managers in electronics industry. Those experts have experience in this industry more than 17 years. First, the criteria list was discussed with four academics and three industry experts. After the experts validated the list, first, fuzzy ANP is applied to find the criteria and subcriteria weights. Second, fuzzy VIKOR is implemented to calculate the scores of smart technologies. The reason to use fuzzy ANP is the capability of its weighting and prioritization ability, and the reason to use fuzzy VIKOR is its ranking ability.

Figure 1 shows the proposed framework. Figure 1 is the answer of the first research question.

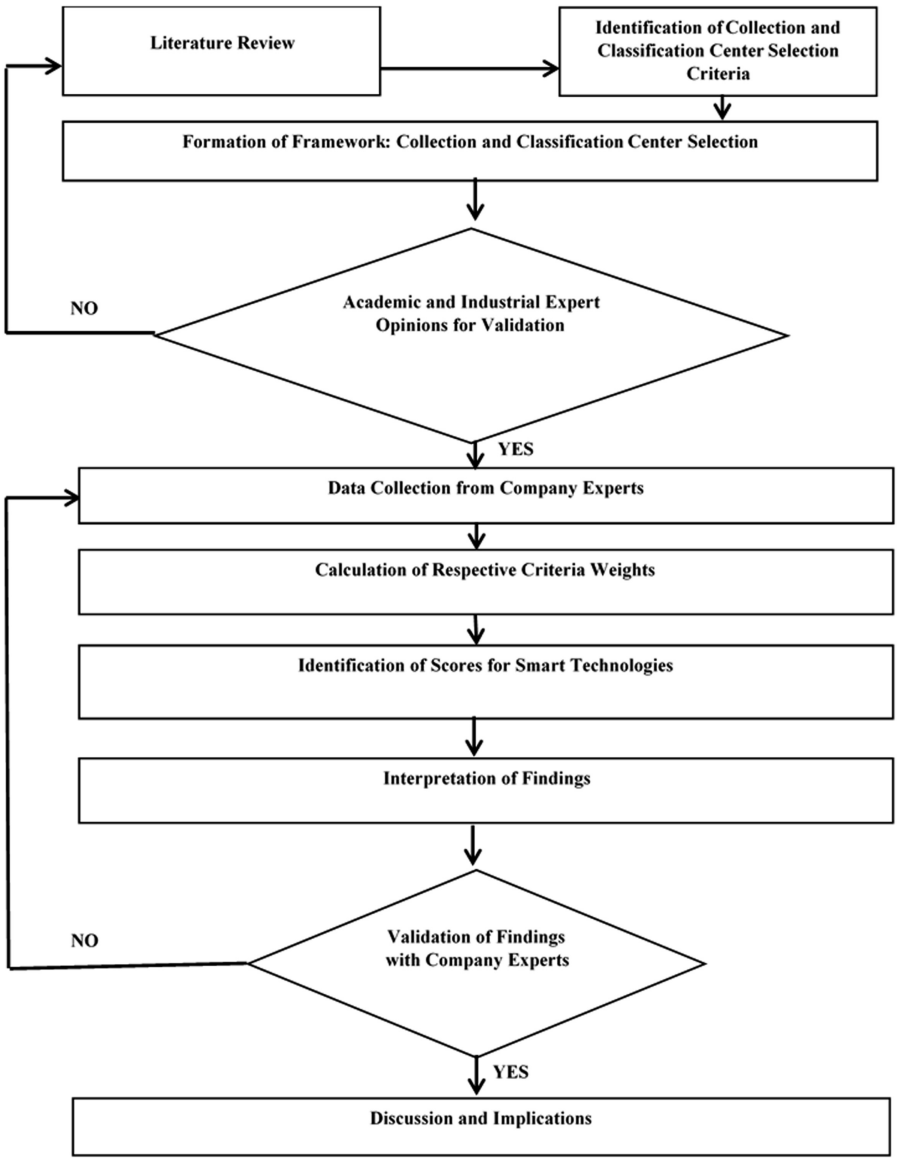


Figure 1.
The proposed framework

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Table 1 shows the criteria list for collection and classification of e-waste. Table 1 is the answer of the second research question.

In the following section, fuzzy ANP and fuzzy VIKOR methods were discussed.

4. Methodology

Using multicriteria decision-making (MCDM) methods is a one of the ways of tackling complex decisions. There are many MCDM methods which may be applied in various areas such as engineering, logistics manufacturing and healthcare decisions (Sagnak *et al.*, 2021). In this study, Fuzzy ANP and fuzzy VIKOR methods were implemented; therefore, in this section, these methods were discussed respectively.

4.1 Fuzzy analytic network process

ANP was first introduced by Saaty (1996). Its capacity to handle qualitative and quantitative factors is its key asset (Sagnak and Kazancoglu, 2019). On the other hand, its usefulness is restricted because of the uncertainty and ambiguity in the nature of decisions (Kazancoglu *et al.*, 2020). As a result, ANP was combined with fuzzy set theory in this study.

The fuzzy version of ANP differs from Saaty's (1996) approach (Sagnak and Kazancoglu, 2019; Kazancoglu *et al.*, 2020). Triangular fuzzy numbers were used for pairwise comparisons. Saaty's (1980) scale is simple to use; nevertheless, when it comes to flexibility and reducing ambiguities in decisions, employing fuzzy numbers rather than crisp values is always a preferable alternative. To determine the decision-makers' preferences, linguistic phrases were employed.

According to Sagnak and Kazancoglu (2019; Yadav and Singh, 2020), Fuzzy ANP involves the following steps:

Step 1: Pairwise comparisons were carried out to find the relationships between criteria.

Step 2: Initial super matrix was created to show the relative significance of cluster k to cluster 1.

Step 3: Weighted super matrix was created by multiplying the first entry of each eigenvector by each entry in the first block of each column, and the second entry with the second block, and so on.

Step 4: The limit super matrix was created by taking power of weighted super matrix until all values for a given row are equal.

Step 5: Normalization procedure is used to get the final crisp weights for each block of the limit super matrix.

4.2 Fuzzy VIKOR

The VIKOR method was first used by Opricovic (1998) and Opricovic and Tzeng (2002). The objective of the VIKOR is to select the best option from a list of potential solutions.

Economic	Social	Environmental
Storage/Holding Cost	Generating Job Opportunities	Proximity to Inhabited Areas
Transportation Cost	Providing Industrial Development	Greenhouse Gas Emissions
Collection Cost	Community Engagement	Pollution Prevention and Control
Energy Cost	Education and Qualification	Effect on Protected Areas
Personnel Cost	Society Benefits	
Operation Cost		

Table 1.
Criteria set

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Because decision-making is inherently ambiguous and subjective, the VIKOR approach is typically used using fuzzy logic (Akram *et al.*, 2021).

The solution stages of fuzzy VIKOR are summarized by Opricovic (2011) as follows:

Stage 1: Find the fuzzy best $\widetilde{f}_j^* = (l_j^*, m_j^*, r_j^*)$ and the fuzzy worst $\widetilde{f}_j^- = (l_j^-, m_j^-, r_j^{-*})$ values by the following formulas, respectively.

$$\widetilde{f}_j^* = \max_i \widetilde{x}_{ij}$$

$$\widetilde{f}_j^- = \min_i \widetilde{x}_{ij}$$

Stage 2: Find the fuzzy difference, $\widetilde{d}_{ij}$ is by:

$$\widetilde{d}_{ij} = \left(\widetilde{f}_j^* - \widetilde{x}_{ij}\right) \Big/ \left(r_j^* - l_j^-\right)$$

Stage 3: Calculate the separation values, $\widetilde{S}_i$, and $\widetilde{R}_i$ of ith alternative by:

$$\widetilde{S}_i = \sum_{j=1}^{m} \widetilde{w}_j \times \widetilde{d}_{ij}$$

$$\widetilde{R}_i = \max_j \widetilde{w}_j \times \widetilde{d}_{ij}$$

where $\widetilde{S}_i = (S_i^l, S_i^m, S_i^r)$ is a weighted sum with respect to the separation value of ith option from f_j^*. $\widetilde{R}_i = (R_i^l, R_i^m, R_i^r)$ identifies the separation value of ith alternative from f_j^-; w_i is the weight of criterion, C_j.

Stage 4: Find the value of $\widetilde{Q}_i = (l_i, m, r_i)$ is by:

$$\widetilde{Q}_i = v\left[\left(\widetilde{S}_i - \widetilde{S}^*\right) \Big/ \left(S^{-r} - S^{*l}\right)\right] + (1-v)\left[\frac{\widetilde{R}_i - \widetilde{R}^*}{R^{-r} - R^{*l}}\right]$$

where $\widetilde{S}^* = min_i \widetilde{S}_i$, $\widetilde{S}^{-r} = max_i \widetilde{S}_i^r$, $\widetilde{R}^* = min_i \widetilde{R}_i$, $\widetilde{R}^{-r} = min_i \widetilde{R}_i$. v refers to the weight for maximum group utility, whereas $(1-v)$ refers to weight for individual regret. The best values of separation values are represented as $\widetilde{S}^*$ and $\widetilde{R}^*$, respectively. Defuzzification is made to calculate the crisp numbers. The best option is given by the minimum Q_i value.

5. Case study

The implementation was conducted in ten companies from the electronics industry in Izmir, Turkey. These companies are creating lots of e-waste. Four of the companies have refurbishing, remanufacturing and reusing policies; but other six do not. All of the companies are operating in the global markets. Some of them are small and medium sized family business and some of them are part of conglomerates.

The main objective is to understand whether smart technologies were beneficial in collection and classification of e-waste. The criteria set includes three main, which are

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economic, social and environmental criteria, including a total of 15 subcriteria. Smart technologies identified in this study were robotics, multiagent systems, autonomous tools, smart vehicles, data-driven technologies, IOT, cloud computing and big data analytics. These smart technologies are the answer of the third research question.

The weights of all criteria were found using fuzzy ANP, and the scores of smart technologies which were useful for collection and classification of e-waste were calculated using fuzzy VIKOR. This is the answer of the fourth research question.

Data collection process includes date gathering through pairwise comparisons consistent with the nature of MCDM methods. Thirty experts from ten companies, which were presented in Table 2, joined the data collection process. These experts, the supply chain managers, supply chain vice managers and sustainability managers, were selected, because the responsibility of collection and classification mechanism is under control of these managers.

The proposed framework and criteria set used in this study are general and usable to similar studies in which smart technologies for collection and classification of e-waste is investigated; but, the findings are unique for this study and cannot be generalized.

Table 3 shows the weights of the main criteria.

Accordingly, the most important criterion for collection and classification of e-waste is found as economic criterion.

Tables 4–6 show the weights of respective subcriteria.

Under the economic cluster, the most important subcriteria were found as collection cost and storage/holding cost, respectively.

Under the social cluster, the most important criteria were found as providing industrial development and society benefits, respectively.

Under the environmental cluster, the most important criteria were found as pollution prevention and control, and greenhouse gas emissions, respectively.

Table 7 shows the overall weights.

Accordingly, in overall manner, the most important criterion was found as collection cost with a weight of 0.167, followed by pollution prevention and control, storage/holding cost and greenhouse gas emissions with weights of 0.115, 0.102 and 0.093, respectively.

Table 8 shows the rankings of smart technologies with respect to criteria weights.

Table 2. Information about experts

Experts	Role	Total experience (Year)	Experts	Role	Total experiences (Year)
1	Supply Chain Manager (SCM)	22	16	SCVM	11
2	SCM	19	17	SCVM	10
3	SCM	18	18	SCVM	13
4	SCM	19	19	SCVM	14
5	SCM	15	20	SCVM	12
6	SCM	17	21	Sustainability Manager (SM)	14
7	SCM	17	22	SM	13
8	SCM	18	23	SM	18
9	SCM	20	24	SM	9
10	SCM	18	25	SM	11
11	Supply Chain Vice Manager (SCVM)	13	26	SM	10
12	SCVM	15	27	SM	10
13	SCVM	14	28	SM	9
14	SCVM	12	29	SM	12
15	SCVM	12	30	SM	11

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Accordingly, autonomous tools were found as the best smart technology for collection and classification of e-waste, followed by robotics and smart vehicles.

6. Discussion & implications

E-waste issue is gaining attention from the literature lately. According to the bibliometric analysis done by Zhang *et al.* (2019b) there has been a rise in research on e-waste in recent years. The scientific literature puts a lot of emphasis on sustainable waste management with the goal of creating a low-carbon economy that can separate economic growth from greenhouse gas emissions. Experts are aware that these actions may present a possibility to lower greenhouse gas emissions (Qu *et al.*, 2013). The scientific community recognizes that waste management may be optimized to achieve benefits for the economy, the environment and social (Cucchiella *et al.*, 2014). In this context, sustainability must address the management of e-waste, and a literature review revealed that there aren't enough operational metrics to quantify and track the effects of resource consumption (Manfredi and Goralczyk, 2013; Nelen *et al.*, 2014). Therefore, it has been demonstrated that even the intention to recycle

Criteria	Weights
Economic	0.478
Social	0.198
Environmental	0.324

Table 3. Weights of main criteria

Subcriteria	Weights
Storage/Holding Cost	0.212
Transportation Cost	0.122
Collection Cost	0.349
Energy Cost	0.128
Personnel Cost	0.090
Operation Cost	0.099

Table 4. Weights of subcriteria under economic criterion

Subcriteria	Weights
Generating Job Opportunities	0.197
Providing Industrial Development	0.297
Community Engagement	0.117
Education and Qualification	0.145
Society Benefits	0.244

Table 5. Weights of subcriteria under-social criterion

Subcriteria	Weights
Proximity to Inhabited Areas	0.177
Greenhouse Gas Emissions	0.287
Pollution Prevention and Control	0.356
Effect on Protected Areas	0.180

Table 6. Weights of subcriteria under-environmental criterion

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Main criteria	SUBCRITERIA	Weights
Economic	Storage/Holding Cost	0.102
	Transportation Cost	0.059
	Collection Cost	0.167
	Energy Cost	0.061
	Personnel Cost	0.043
	Operation Cost	0.047
Social	Generating Job Opportunities	0.039
	Providing Industrial Development	0.059
	Community Engagement	0.023
	Education and Qualification	0.029
	Society Benefits	0.048
Environmental	Proximity to Inhabited Areas	0.057
	Greenhouse Gas Emissions	0.093
	Pollution Prevention and Control	0.115
	Effect on Protected Areas	0.058

Table 7.
Overall weights

Order	Smart technologies	Qj
1	Autonomous Tools	0
2	Robotics	0.046873
3	Smart Vehicles	0.13186
4	Multiagent Systems	0.546297
5	Big Data Analytics	0.551391
6	Data-Driven Technologies	0.760544
7	IOT	0.937999
8	Cloud Computing	1

Table 8.
Rankings of smart technologies

e-waste is influenced by a variety of factors, including gender, marital status, residential conditions, economic benefits, knowledge of the potential toxicity of e-waste, perceptions of recycling, social norms, the convenience of recycling facilities and services, and others (Wang *et al.*, 2011, 2016; Saphores *et al.*, 2012; Otto *et al.*, 2018).

The reason why e-waste gains so much attention from both scholars, policymakers and sectors is because of the constantly changing consumption habits brought by the developing technology, the short product life cycle of the electronic products and the limited repair or improvement possibilities of the electronics, the harms of e-waste to the environment, economy and social aspects. E-waste is a major issue today because it is times-consuming and expensive to collect these products, the disassembly processes are complicated (there are too many materials and components, different and diverse classifications, and difficult separation), the usage of uncontrolled and unhealthy landfills, inappropriate recycling techniques, harmful disposal techniques and informal recycling. Moreover, these wastes contain a variety of substances, including important, valuable and hazardous ones, that call for a special recycling procedure to prevent problems with the environment and human health as well as the environmental costs associated with the extraction and refinement of primary new materials. The majority of e-waste is destroyed using crude recycling methods, which include open burning of plastics (to reduce waste volume), copper wires (to recover valuable metals, such as copper), leaching printed wiring boards (PWBs) in strong acids to recover precious metals (the waste acid from which was discharged into nearby streams) and grilling PWBs over honeycomb-coal fires to melt solder (to allow the collection of electronic

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components, e.g. diodes and resistors) (Liu *et al.*, 2006; Wong *et al.*, 2007; Nnorom and Osibanjo, 2008).

When these electronic items are stored in landfills, many dangerous metals (such as mercury, lead, cadmium, iron and aluminum) combine with the soil and water, inflicting harm to the ecosystem that we are unable to repair. These wastes contain a variety of substances, including important, valuable and hazardous ones, that call for a special recycling procedure to prevent problems with the environment and human health as well as the environmental costs associated with the extraction and refinement of primary new materials. In fact, the sources of e-waste are prohibited from dumping waste because it includes dangerous elements, according to a number of e-waste recycling legislation. However, unauthorized dumping by sources is a significant issue (Wakolbinger *et al.*, 2014).

Especially, these landfills serve as a significant supply of secondary raw materials, and many nonferrous elements (such as gold, silver, platinum, etc.) are referred to as recoverable. Due to untrained workers, precious metal recovery yields are extremely low in impoverished nations (Chatterjee and Kumar, 2009); in industrialized nations, recovery rates for silver, palladium and gold are greater than 90% but not quite 100% (Park and Fray, 2009; Wakolbinger *et al.*, 2014). Also, without the proper handling, the enormous amount of e-waste, which contains a variety of harmful elements and components, might pose a serious challenge to sustainability (Hicks *et al.*, 2005; Kahhat and Williams, 2012; Qu *et al.*, 2013). Hence, the rate of e-waste collection must be increased in order to reduce negative environmental effects (Reck and Graedel, 2012; Taghipour *et al.*, 2012; Qu *et al.*, 2013). However, collection, storing & handling, and discarding activities of this e-waste is expensive and difficult to perform. Even when discarding activities incur expenses, as is commonly the case with recycling product materials, holding costs functions as a disincentive to discard the goods (Wakolbinger *et al.*, 2014.) Because, e-waste, which has complex processes such as the collection of materials, separation by disassembly, identification and classification of materials and components, is also harmful and deadly for human health in terms of the situations that arise during these processes. Typically, these operations don't use much, if any, personal protective equipment or pollution control measures (Nnorom and Osibanjo, 2008). Thereby, inadequate handling can result in workplace dangers, serious environmental contamination and value loss from recyclable and reused materials (Batoo *et al.*, 2022). In addition, these processes can be damaged by human error. In this context, the fact that these processes can be carried out by autonomous tools and robots that minimize the human contribution and carry out more power-requiring and dangerous tasks (such as toxic emissions during the process). By doing that, autonomous tools and robotics can provide many benefits towards sustainability in terms of social (decreased exposure of employees to harmful gases), economic (eliminated collection, storage & handling cost), and environmental (such as minimized greenhouse gas emission, mixing of harmful metals into the soil and water). E-waste is considered the most difficult type of waste because of this.

Digitalization includes new developments in industries that help society, the economy and the environment by lowering air pollution, raising working standards, cutting waste and other things through new technologies like robotics, sensors and smart vehicles (Abdel-Shafy and Mansour, 2018). Therefore, by utilizing these cutting-edge technologies like robotics, sensors and smart vehicles, the digitalization process benefits society, business, industry and the environment by reducing waste, raising working standards and reducing air pollution (Baldé *et al.*, 2017; Batoo *et al.*, 2022). Our findings suggest that autonomous tools and robotics are one of the most promising technologies in the e-waste problem which are inventive and safer modes of transportation when taking environmental issues into account (Lopez-Lambas and Alonso, 2019; Hasan *et al.*, 2020). In contrast to numerous communal trash cans and piles of bagged street garbage, robotic bins, which are more frequently used in waste collection systems, offer a fully automated system that sorts and

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separates e-waste through reverse logistics activities (Baldé *et al.*, 2017). For an autonomous disassembly procedure, Büker *et al.* (2001) use a vision-guided robotic system.

In addition, another important technology that the results of this study show smart vehicles are substantial for achieving the sustainability of the e-waste system. Despite some research (Park *et al.*, 2017) arguing that the way of collecting e-waste in emerging economies can help to reduce collection costs, however, the methods of collection are hazardous to human health (Gu *et al.*, 2016). In this sense, Esmaeilian *et al.* (2018) asserted that smart vehicle systems offer better monitoring and data sharing in e-waste collection, this is in line with their contributions. The smart vehicle provides financial advantages by lowering transportation costs while enabling controlling, data sharing and control in the e-waste process (Batoo *et al.*, 2022). In addition to that their many advantages that smart vehicles provide for major e-waste problems. For instance, smart vehicles offer advantages including time optimization, reduced labor expenses and flexibility in modifying routes in the system for collecting e-waste. Additionally, driverless vehicles give businesses the ability to cut labor expenses, guarantee time efficiency and provide safe transportation for the collection of e-waste (Sagnak *et al.*, 2021).

The main contribution of this study is to examine and assess the most serious issues that e-waste is confronting in terms of sustainability (economic, social and environmental) and propose a solution to these issues by integrating digitalization into the e-waste system to minimize the detrimental effects of the current system. In this context, the most promising technologies were analyzed for predetermined issues of e-waste and related implications were presented by using these technologies to overcome the found serious issues in e-waste.

7. Conclusion

Waste management is one of the most important issue especially for developing countries where sustainability problems are at critical levels. Among the different types of waste, the most important is e-waste.

There is a significant gap in the past studies with respect to the integration of smart technologies with e-waste collection and classification mechanism (Rada *et al.*, 2013). Adaptation of smart technologies has become important in the e-waste operations (Kazancoglu *et al.*, 2021). Therefore, this study presents a framework integrating the concepts of collection and classification mechanisms and smart technologies. The criteria set was developed for the different characteristics of collection and classification centers, and the smart technologies were identified as the solutions for collection and classification mechanism. The criteria set includes three main, which are economic, social and environmental criteria, including a total of 15 subcriteria. Smart technologies identified in this study were robotics, multiagent systems, autonomous tools, smart vehicles, data-driven technologies, IOT, cloud computing and big data analytics. The weights of all criteria were found using fuzzy ANP, and the scores of smart technologies which were useful for collection and classification of e-waste were calculated using fuzzy VIKOR.

Accordingly, in overall manner, the most important criterion was found as collection cost with a weight of 0.167, followed by pollution prevention and control, storage/holding cost and greenhouse gas emissions with weights of 0.115, 0.102, and 0.093, respectively. Also, autonomous tools were found as the best smart technology for collection and classification of e-waste, followed by robotics and smart vehicles.

The implementation of the study was conducted in an emerging economy, Turkey, which can be determined as the limitation of the study. Another limitation is that all numerical calculations are based on the subjective judgments of decision-makers, and therefore, the findings of the study cannot be generalized.

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Further possible studies may focus on the application of the proposed framework in other emerging economies. The possible future studies may focus on using different MCDM methods to compare the results. Also, some other new smart technologies may be determined to solve the collection and classification of e-waste.

References

Abdel-Shafy, H.I. and Mansour, M.S. (2018), "Solid waste issue: sources, composition, disposal, recycling, and valorization", *Egyptian Journal of Petroleum*, Vol. 27 No. 4, pp. 1275-1290.

Ahmad, F., Jamal, A., Mazher, K.M., Umer, W. and Iqbal, M. (2021), "Performance evaluation of plastic concrete modified with e-waste plastic as a partial replacement of coarse aggregate", *Materials*, Vol. 15 No. 1, pp. 175-193.

Akram, M., Kahraman, C. and Zahid, K. (2021), "Group decision-making based on complex spherical fuzzy VIKOR approach", *Knowledge-Based Systems*, Vol. 216, 106793.

Alcayaga, A., Wiener, M. and Hansen, E.G. (2019), "Towards a framework of smart-circular systems: an integrative literature review", *Journal of Cleaner Production*, Vol. 221, pp. 622-634.

Anagnostopoulos, T., Zaslavsky, A., Kolomvatsos, K., Medvedev, A. and Amirian, P. (2017), "Challenges and opportunities of waste management in IoT-enabled smart cities: a survey", *IEEE Transactions on Sustainable Computing*, Vol. 2 No. 3, pp. 275-289.

Archibong Bassey, B. and Roseline William, A. (2020), "The arts of converting waste to wealth: towards environmental sustainability in Nigeria", *Central Asian Journal of Environmental Science and Technology Innovation*, Vol. 1 No. 3, pp. 159-167.

Ashwin, M., Alqahtani, A.S. and Mubarakali, A. (2021), "IOT based intelligent route selection of wastage segregation for smart cities using solar energy", *Sustainable Energy Technologies and Assessments*, Vol. 46, 101281.

Baldé, C.P., Forti, V., Gray, V., Kuehr, R. and Stegmann, P. (2017), *The Global E-Waste Monitor 2017: Quantities, Flows and Resources, United Nations University (UNU)*, International Telecommunication Union (ITU) & International Solid Waste Association (ISWA), Bonn/Geneva/Vienna.

Batoo, K.M., Pandiaraj, S., Muthuramamoorthy, M., Raslan, E.H. and Krishnamoorthy, S. (2022), "Behavior-based swarm model using fuzzy controller for route planning and E-waste collection", *Environmental Science and Pollution Research*, Vol. 29 No. 14, pp. 19940-19954.

Bjorn, A., Chandrakumar, C., Boulay, A.M., Doka, G., Fang, K., Gondran, N. and Ryberg, M. (2020), "Review of life-cycle based methods for absolute environmental sustainability assessment and their applications", *Environmental Research Letters*, Vol. 15 No. 8, 083001.

Borghi, A.D., Gallo, M., Strazza, C., Magrassi, F. and Castagna, M. (2014), "Waste management in Smart Cities: the application of circular economy in Genoa (Italy)", *Impresa Progetto Electronic Journal of Management*, Vol. 4, pp. 1-13.

Broo, D.G., Lamb, K., Ehwi, R.J., Pärn, E., Koronaki, A., Makri, C. and Zomer, T. (2021), "Built environment of Britain in 2040: scenarios and strategies", *Sustainable Cities and Society*, Vol. 65, 102645.

Büker, U., Drüe, S., Götze, N., Hartmann, G., Kalkreuter, B., Stemmer, R. and Trapp, R. (2001), "Vision-based control of an autonomous disassembly station", *Robotics and Autonomous Systems*, Vol. 35 Nos 3-4, pp. 179-189.

Chandra, P., Yeh, C.H. and Dutta, P. (2022), "Managing strategies for mitigating interacting barriers to sustainable online e-waste collection platforms in India", *Benchmarking: An International Journal*, Vol. ahead-of-print No. ahead-of-print, doi: 10.1108/BIJ-12-2021-0776.

Chatterjee, S. and Kumar, K. (2009), "Effective electronic waste management and recycling process involving formal and non-formal sectors", *International Journal of Physical Sciences*, Vol. 4 No. 13, pp. 893-905.

Downloaded from www.emerald.com/ijqrm/article-pdf/41/9/2434/9522692/ijqrm-08-2022-0259.pdf by Amrita Vishwa Vidyapeetham user on 10 October 2026

Cong, W., Jiongming, Q., Cheng, Q., Xu, R., Chuanjun, L. and Bin, C. (2021), "A smart municipal waste management system based on deep-learning and Internet of Things", *Waste Management*, Vol. 135, pp. 20-29.

Copani, G., Picone, N., Colledani, M., Pepe, M. and Tasora, A. (2019), "Highly evolvable E-waste recycling technologies and systems", in Tolio, T., Copani, G. and Terkaj, W. (Eds), *Factories of the Future*, Springer, Cham, pp. 109-128.

Cucchiella, F., D'Adamo, I. and Gastaldi, M. (2014), "Sustainable management of waste-to-energy facilities", *Renewable and Sustainable Energy Reviews*, Vol. 33, pp. 719-728.

Dave, S.R., Shah, M.B. and Tipre, D.R. (2016), "E-waste: metal pollution threat or metal resource", *Journal of Advanced Research in Biotechnology*, Vol. 1 No. 2, pp. 1-14.

Esmaeilian, B., Wang, B., Lewis, K., Duarte, F., Ratti, C. and Behdad, S. (2018), "The future of waste management in smart and sustainable cities: a review and concept paper", *Waste Management*, Vol. 81, pp. 177-195.

Fatimah, Y.A., Govindan, K., Murniningsih, R. and Setiawan, A. (2020), "Industry 4.0 based sustainable circular economy approach for smart waste management system to achieve sustainable development goals: a case study of Indonesia", *Journal of Cleaner Production*, Vol. 269, 122263.

Feenberg, A. (2019), "The internet as network, world, co-construction, and mode of governance", *The Information Society*, Vol. 35 No. 4, pp. 229-243.

Fuss, M., Barros, R.T.V. and Poganietz, W.R. (2018), "Designing a framework for municipal solid waste management towards sustainability in emerging economy countries-an application to a case study in Belo Horizonte (Brazil)", *Journal of Cleaner Production*, Vol. 178, pp. 655-664.

Garlapati, V.K. (2016), "E-waste in India and developed countries: management, recycling, business and biotechnological initiatives", *Renewable and Sustainable Energy Reviews*, Vol. 54, pp. 874-881.

Gu, Y., Wu, Y., Wang, H. and Zuo, T. (2016), "The stability and profitability of the informal WEEE collector in developing countries: a case study of China", *Resources, Conservation and Recycling*, Vol. 107, pp. 18-26.

Hasan, U., Whyte, A. and Al Jassmi, H. (2020), "A review of the transformation of road transport systems: are we ready for the next step in artificially intelligent sustainable Transport?", *Applied System Innovation*, Vol. 3 No. 1, pp. 1-21.

Heeks, R., Subramanian, L. and Jones, C. (2015), "Understanding e-waste management in developing countries: strategies, determinants, and policy implications in the Indian ICT sector", *Information Technology for Development*, Vol. 21 No. 4, pp. 653-667.

Hicks, C., Dietmar, R. and Eugster, M. (2005), "The recycling and disposal of electrical and electronic waste in China - legislative and market responses", *Environmental Impact Assessment Review*, Vol. 25 No. 5, pp. 459-471.

Hoque, M.M. and Rahman, M.T.U. (2020), "Landfill area estimation based on solid waste collection prediction using ANN model and final waste disposal options", *Journal of Cleaner Production*, Vol. 256, 120387.

Ikhlayel, M. (2018), "An integrated approach to establish e-waste management systems for developing countries", *Journal of Cleaner Production*, Vol. 170, pp. 119-130.

Imran, M., Haydar, S., Kim, J., Awan, M.R. and Bhatti, A.A. (2017), "E-waste flows, resource recovery and improvement of legal framework in Pakistan", *Resources, Conservation and Recycling*, Vol. 125, pp. 131-138.

Jahangir, M.H., Mokhtari, R. and Mousavi, S.A. (2021), "Performance evaluation and financial analysis of applying hybrid renewable systems in cooling unit of data centers–A case study", *Sustainable Energy Technologies and Assessments*, Vol. 46, 101220.

Joensuu, T., Edelman, H. and Saari, A. (2020), "Circular economy practices in the built environment", *Journal of Cleaner Production*, Vol. 276, 124215.

Downloaded from www.emerald.com/ijqrm/article-pdf/41/9/2434/9522692/ijqrm-08-2022-0259.pdf by Amrita Vishwa Vidyapeetham user on 10 October 2026

Kahhat, R. and Williams, E. (2012), "Materials flow analysis of e-waste: domestic flows and exports of used computers from the United States", *Resources, Conservation and Recycling*, Vol. 67, pp. 67-74.

Kazancoglu, Y., Sagnak, M., Kayikci, Y. and Mangla, S.K. (2020), "Operational excellence in a green supply chain for environmental management – a case study", *Business Strategy and the Environment*, Vol. 29 No. 3, pp. 1532-1547.

Kazancoglu, Y., Ozbiltekin, M., Ozen, Y.D.O. and Sagnak, M. (2021), "A proposed sustainable and digital collection and classification center model to manage e-waste in emerging economies", *Journal of Enterprise Information Management*, Vol. 34 No. 1, pp. 267-291.

Keim, R.G. (2017), "The future is now", *Journal of Clinical Orthodontics*, Vol. 51 No. 1, pp. 9-10.

Kumar, A., Holuszko, M. and Espinosa, D.C.R. (2017), "E-Waste: an overview on generation, collection, legislation and recycling practices", *Resources, Conservation and Recycling*, Vol. 122, pp. 32-42.

Liu, X., Tanaka, M. and Matsui, Y. (2006), "Electrical and electronic waste management in China: progress and the barriers to overcome", *Waste Management and Research*, Vol. 24 No. 1, pp. 92-101.

López Alvarez, J.V., Aguilar Larrucea, M., Fernández-Carrión Quero, S. and Jiménez del Valle, A. (2008), "Optimizing the collection of used paper from small businesses through GIS techniques: the Leganés case (Madrid, Spain)", *Waste Management*, Vol. 28 No. 2, pp. 282-293.

López-Lambas, M.E. and Alonso, A. (2019), "The driverless bus: an analysis of public perceptions and acceptability", *Sustainability*, Vol. 11 No. 18, pp. 4986-5000.

Lu, S.Y., Li, Y.X., Zhang, T., Cai, D., Ruan, J.J., Huang, M.Z. and Qiu, R.L. (2017), "Effect of e-waste recycling on urinary metabolites of organophosphate flame retardants and plasticizers and their association with oxidative stress", *Environmental Science and Technology*, Vol. 51 No. 4, pp. 2427-2437.

Manfredi, S. and Goralczyk, M. (2013), "Life cycle indicators for monitoring the environmental performance of European waste management", *Resources, Conservation and Recycling*, Vol. 81, pp. 8-16.

Martin, C. and Leurent, H. (2017), *Technology and Innovation for the Future of Production: Accelerating Value Creation*, World Economic Forum, Geneva, pp. 1-38.

Mohammadi, E., Singh, S.J. and Habib, K. (2021), "Electronic waste in the Caribbean: an impending environmental disaster or an opportunity for a circular economy?", *Resources, Conservation and Recycling*, Vol. 164, 105106.

Mundada, M.N., Kumar, S. and Shekdar, A.V. (2004), "E-waste: a new challenge for waste management in India", *International Journal of Environmental Studies*, Vol. 61, pp. 265-279.

Murugesan, A., Loganathan, M., Kumar, P.S. and Vo, D.V.N. (2021), "Cobalt and nickel oxides supported activated carbon as an effective photocatalysts for the degradation Methylene Blue dye from aquatic environment", *Sustainable Chemistry and Pharmacy*, Vol. 21, 100406.

Nelen, D., Manshoven, S., Peeters, J.R., Vanegas, P., D'Haese, N. and Vrancken, K. (2014), "A multidimensional indicator set to assess the benefits of WEEE material recycling", *Journal of Cleaner Production*, Vol. 83, pp. 305-316.

Nnorom, I.C. and Osibanjo, O. (2008), "Overview of electronic waste (e-waste) management practices and legislations, and their poor applications in the developing countries", *Resources, Conservation and Recycling*, Vol. 52 No. 6, pp. 843-858.

Opricovic, S. (1998), *Multi-criteria Optimization of Civil Engineering Systems*, Faculty of Civil Engineering, Belgrade.

Opricovic, S. (2011), "Fuzzy VIKOR with an application to water resources planning", *Expert Systems with Applications*, Vol. 38 No. 10, pp. 12983-12990.

Opricovic, S. and Tzeng, G.H. (2002), "Multicriteria planning of post-earthquake sustainable reconstruction", *Computer-Aided Civil and Infrastructure Engineering*, Vol. 17 No. 3, pp. 211-220.

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Osburg, T. and Lohrmann, C. (2017), *Sustainability in a Digital World: New Opportunities through New Technologies*, Springer, New York.

Otto, S., Kibbe, A., Henn, L., Hentschke, L. and Kaiser, F.G. (2018), "The economy of E-waste collection at the individual level: a practice oriented approach of categorizing determinants of E-waste collection into behavioral costs and motivation", *Journal of Cleaner Production*, Vol. 204, pp. 33-40.

Ozturk, T. (2015), "Generation and management of electrical–electronic waste (E-Waste) in Turkey", *Journal of Material Cycles and Waste Management*, Vol. 17 No. 3, pp. 411-421.

Park, Y.J. and Fray, D.J. (2009), "Recovery of high purity precious metals from printed circuit boards", *Journal of Hazardous Materials*, Vol. 164 Nos 2-3, pp. 1152-1158.

Park, J.K., Hoerning, L., Watry, S., Burgett, T. and Matthias, S. (2017), "Effects of electronic waste on developing countries", *Advance Recycling Waste Management*, Vol. 2 No. 2, pp. 1-6.

Qu, Y., Zhu, Q., Sarkis, J., Geng, Y. and Zhong, Y. (2013), "A review of developing an e-wastes collection system in Dalian, China", *Journal of Cleaner Production*, Vol. 52, pp. 176-184.

Rada, E.C., Ragazzi, M. and Fedrizzi, P. (2013), "Web-GIS oriented systems viability for municipal solid waste selective collection optimization in developed and transient economies", *Waste Management*, Vol. 33 No. 4, pp. 785-792.

Reck, B.K. and Graedel, T.E. (2012), "Challenges in metal recycling", *Science*, Vol. 337 No. 6095, pp. 690-695.

Robinson, B.H. (2009), "E-waste: an assessment of global production and environmental impacts", *Science of the Total Environment*, Vol. 408 No. 2, pp. 183-191.

Rutqvist, D., Kleyko, D. and Blomstedt, F. (2019), "An automated machine learn- ing approach for smart waste management systems", *IEEE Transactions on Industrial Informatics*, Vol. 16 No. 1, pp. 384-392.

Saaty, T. (1980), *The Analytic Hierarchy Process*, 1st ed., RWS Publications McGraw-Hill, Pittsburgh, PA.

Saaty, T. (1996), *Decision Making with Dependence and Feedback*, RWS Publications, Pittsburgh, PA.

Sagnak, M. and Kazancoglu, Y. (2019), "Integrated fuzzy analytic network process and 0-1 goal programming technique for enterprise resource planning (ERP) software selection", *Ege Academic Review*, Vol. 19 No. 1, pp. 75-88.

Sagnak, M., Berberoglu, Y., Memis, I. and Yazgan, O. (2021), "Sustainable collection center location selection in emerging economy for electronic waste with fuzzy Best-Worst and fuzzy TOPSIS", *Waste Management*, Vol. 127, pp. 37-47.

Sahajwalla, V. and Hossain, R. (2020), "The science of microrecycling: a review of selective synthesis of materials from electronic waste", *Materials Today Sustainability*, Vol. 9, 100040.

Saphores, J.D.M., Ogunseitan, O.A. and Shapiro, A.A. (2012), "Willingness to engage in a pro-environmental behavior: an analysis of e-waste recycling based on a national survey of US households", *Resources, Conservation and Recycling*, Vol. 60, pp. 49-63.

Shahabuddin, M., Uddin, M.N., Chowdhury, J.I., Ahmed, S.F., Uddin, M.N., Mofijur, M. and Uddin, M.A. (2022), "A review of the recent development, challenges, and opportunities of electronic waste (e-waste)", *International Journal of Environmental Science and Technology*, Vol. ahead-of-print No. ahead-of-print, doi: 10.1007/s13762-022-04274-w.

Sharma, M., Joshi, S., Kannan, D., Govindan, K., Singh, R. and Purohit, H.C. (2020), "Internet of Things (IoT) adoption barriers of smart cities' waste management: an Indian context", *Journal of Cleaner Production*, Vol. 270, 122047.

Shevchenko, T., Saidani, M., Danko, Y., Golysheva, I., Chovancová, J. and Vavrek, R. (2021), "Towards a smart E-waste system utilizing supply chain participants and interactive online maps", *Recycling*, Vol. 6 No. 1, pp. 8-21.

Shittu, O.S., Williams, I.D. and Shaw, P.J. (2021), "Global E-waste management: can WEEE make a difference? A review of e-waste trends, legislation, contemporary issues and future challenges", *Waste Management*, Vol. 120, pp. 549-563.

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Shreyas Madhav, A.V., Rajaraman, R., Harini, S. and Kiliroor, C.C. (2022), "Application of artificial intelligence to enhance collection of E-waste: a potential solution for household WEEE collection and segregation in India", *Waste Management and Research*, Vol. 40 No. 7, pp. 1047-1053.

Singh, S., Dasgupta, M.S. and Routroy, S. (2022), "Analysis of critical success factors to design E-waste collection policy in India: a fuzzy DEMATEL approach", *Environmental Science and Pollution Research*, Vol. 29, pp. 10585-10604.

Taghipour, H., Nowrouz, P., Jafarabadi, M.A., Nazari, J., Hashemi, A.A., Mosaferi, M. and Dehghanzadeh, R. (2012), "E-waste management challenges in Iran: presenting some strategies for improvement of current conditions", *Waste Management and Research*, Vol. 30 No. 11, pp. 1138-1144.

Wakolbinger, T., Toyasaki, F., Nowak, T. and Nagurney, A. (2014), "When and for whom would e-waste be a treasure trove? Insights from a network equilibrium model of e-waste flows", *International Journal of Production Economics*, Vol. 154, pp. 263-273.

Wang, Z., Zhang, B., Yin, J. and Zhang, X. (2011), "Willingness and behavior towards e-waste recycling for residents in Beijing city, China", *Journal of Cleaner Production*, Vol. 19 Nos 9-10, pp. 977-984.

Wang, Z., Guo, D. and Wang, X. (2016), "Determinants of residents' e-waste recycling behaviour intentions: evidence from China", *Journal of Cleaner Production*, Vol. 137, pp. 850-860.

Wang, F., Yu, L. and Wu, A. (2021), "Forecasting the electronic waste quantity with a decomposition-ensemble approach", *Waste Management*, Vol. 120, pp. 828-838.

Widmer, R., Oswald-Krapf, H., Sinha-Khetriwal, D., Schnellmann, M. and Böni, H. (2005), "Global perspectives on e-waste", *Environmental Impact Assessment Review*, Vol. 25, pp. 436-458.

Wong, C.S., Wu, S.C., Duzgoren-Aydin, N.S., Aydin, A. and Wong, M.H. (2007), "Trace metal contamination of sediments in an e-waste processing village in China", *Environmental Pollution*, Vol. 145 No. 2, pp. 434-442.

Xia, W., Jiang, Y., Chen, X. and Zhao, R. (2021), "Application of machine learning algorithms in municipal solid waste management: a mini review", *Waste Management and Research*, Vol. 40 No. 6, pp. 609-624.

Yadav, S. and Singh, S.P. (2020), "An integrated fuzzy-ANP and fuzzy-ISM approach using blockchain for sustainable supply chain", *Journal of Enterprise Information Management*, Vol. 34 No. 1, pp. 54-78.

Yukalang, N., Clarke, B. and Ross, K. (2017), "Barriers to effective municipal solid waste management in a rapidly urbanizing area in Thailand", *International Journal of Environmental Research and Public Health*, Vol. 14 No. 9, pp. 9-14.

Zhang, A., Venkatesh, V.G., Liu, Y., Wan, M., Qu, T. and Donald, H. (2019a), "Barriers to smart waste management for a circular economy in China", *Journal of Cleaner Production*, Vol. 240, 118198.

Zhang, L., Geng, Y., Zhong, Y., Dong, H. and Liu, Z. (2019b), "A bibliometric analysis on waste electrical and electronic equipment research", *Environmental Science and Pollution Research*, Vol. 26 No. 21, pp. 21098-21108.

Zhou, Y., Sun, J., Wang, L., Zhu, G., Li, M., Liu, J. and Yin, G. (2021), "Multiple classes of chemical contaminants in soil from an e-waste disposal site in China: occurrence and spatial distribution", *Science of the Total Environment*, Vol. 752, 141924.

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